MINI PROJECT REPORT

ON

“DAIRY MANAGEMENT SYSTEM”

Submitted for the partial fulfillment of the Requirement for the award of degree Bachelor of Computer Applications

UTTARANCHAL UNIVERSITY DEHRADUN | NAAC GRADE A+

(Session: 2025-2026)

Under the supervision of

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ACKNOWLEDGEMENT

The most awaited moment of any endeavor is successful completion but nothing can be done successfully if done alone. Success is the outcome of contribution and consistent help of various persons and we thank those ones who helped us in successful completion of this project.

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Rashu Malik
B.C.A 2nd Sem
University Roll No. 2509000327

DECLARATION

I hereby declare that the project report entitled **Dairy Management System** submitted by **Rashu** to Uttaranchal School of Computing Sciences. The project report was done under the guidance of **Mr. Deepak kumar, Assistant Professor-USCS**.
I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university or institute.

Rashu Malik
B.C.A 2nd Sem
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CERTIFICATE OF ORIGINALITY

This is to certify **Rashu Malik** student of BCA 2nd semester, of **Uttaranchal School of Computing Sciences, Dehradun**, has completed the Project Report for the topic **Dairy Management System** for the session (2025 – 2026).

Under the guidance of:

Mr. Deepak kumar

Assistant Professor-USCS

Uttaranchal University, Dehradun

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INTRODUCTION

We are creating a Window application for “**Dairy management system**”.

In C++ language by using Code block & Dev C++. The mission of the Milk Production House Project is to create to communication between rural area people and dairy management. Our goal is to develop this application to encourage our dairy industry. Dairy milk management system is a software application to maintain day to day transactions in a Milk dairy. This software helps to register all the suppliers, Buyer details, purchase, Sales details. Our plan is to collect the Milk from many different sources and distribute the same to the Milk buyers.

Objectives:

To develop a Dairy Management System for managing daily dairy operations.
To maintain records of milk purchase, sales, suppliers, and customers.
To reduce manual work and human errors through automation.
To generate bills and maintain transaction records efficiently.
To improve the overall efficiency and management of dairy businesses.

PROFILES OF TWO IT COMPANIES

Tata Consultancy Services (TCS)

Tata Consultancy Services (TCS) is one of the world's largest and most trusted Information Technology (IT) services, consulting, and business solutions companies. Established in **1968**, TCS is a flagship company of the **Tata Group**, India's most respected business conglomerate. Its headquarters is located in **Mumbai**, and the company operates across more than **150 locations in 46+ countries**, serving global clients across diverse industries such as banking, financial services, insurance, healthcare, retail, telecom, manufacturing, energy, and government sectors. Over the years, TCS has built a strong reputation for delivering high-quality IT solutions, business transformation services, and advanced digital technologies. The company is valued worldwide for its research-driven approach, mainly through **TCS Innovation Labs**, which work on cutting-edge technologies and patents. This innovation culture has helped TCS consistently rank among the world's most valuable IT services brands.

TCS offers a broad portfolio of services and solutions that cater to global business needs. Its **IT Services & Consulting** division provides end-to-end strategic and technical support. In **Application Development and Maintenance**, TCS designs, builds, and manages large-scale enterprise applications that support business operations. It is also a major provider of **Cloud Computing services**, helping organizations migrate to leading cloud platforms such as AWS, Microsoft Azure, and Google Cloud. TCS delivers AI-powered solutions, including predictive analytics, automation tools, and chatbots, enabling companies to adopt intelligent systems. In the field of **Cybersecurity**, TCS provides essential services like threat detection, identity management, and risk assessment to protect organizations from cyber threats. The company also specializes in **Enterprise Resource Planning (ERP)** solutions such as SAP and Oracle, helping industries streamline operations. Additionally, TCS runs extensive **Business Process Outsourcing (BPO)** services such as customer support and financial processing, enabling companies to reduce costs and improve operational efficiency.

The company has deep expertise in modern and emerging technologies. It works with a wide set of **programming languages**, including Java, Python, C++, and C#. Its teams are skilled in advanced **frameworks and platforms** like Spring Boot, .NET, React, Angular, and Node.js. TCS is also a major player in cloud technologies, with certified experts on AWS, Azure, and Google Cloud. Its database capabilities include SQL-based and NoSQL systems such as Oracle, MySQL, and MongoDB. Beyond traditional technologies, TCS invests heavily in advanced fields like **Blockchain**, **IoT (Internet of Things)**, **Robotic Process Automation (RPA)**, and **Big Data Analytics** using tools such as Hadoop and Spark. The company's internal learning platforms ensure that employees continuously upgrade their skills to keep pace with evolving industry demands.

TCS is equally known for its strong and employee-friendly **work culture**. It offers excellent training programs for fresh graduates, including TCS Xplore, which prepares newcomers for real-world projects. Employees enjoy a healthy work-life balance and opportunities to work on international projects in countries like the USA, UK, Europe, and Australia. The company encourages teamwork, creativity, innovation, and continuous learning. With various internal certification programs and leadership development tracks, TCS provides ample opportunities for career growth.

The company's **placement process** is structured and transparent. For eligibility, students must have a minimum of **60% marks** throughout their academic history and must not have any active backlogs. The selection process usually includes four rounds: an online aptitude and logical reasoning test, a technical interview focusing on programming and problem-solving skills, a managerial interview in some cases, and finally an HR interview to assess communication skills and personality. This multi-step process ensures that the company recruits candidates with strong analytical ability, technical knowledge, and professional behavior.

For communication, TCS can be contacted through email at **corporate.tcs@tcs.com**, phone at **+91-22-6778 9999**, or via its official LinkedIn page.

Infosys Limited

Infosys Limited is a leading global information technology, consulting, and outsourcing company founded in **1981** and headquartered in **Bengaluru, India**. Operating in more than **50 countries**, Infosys helps organizations transform into digital and data-driven enterprises. The company is widely known for its strong focus on innovation, sustainability, and advanced technology research. Its **Global Education Center in Mysore** is one of the world's largest corporate training campuses, providing high-quality technical and professional training to freshers. Initiatives like **Infosys Labs** and the AI platform **Infosys Nia** strengthen its position in digital intelligence and automation.
Infosys offers a diverse range of services, including **consulting**, **software development**, **maintenance**, **cloud and infrastructure services**, and **cybersecurity**. The company specializes in digital transformation through **AI, ML, automation**, and **UI/UX design**. Additionally, Infosys implements enterprise systems such as **SAP**, **Oracle**, and **CRM platforms**, helping businesses improve efficiency and adopt modern technologies.
The company works with a wide range of technologies, including programming languages like **Python, Java, JavaScript, Kotlin, and Swift**, and front-end frameworks such as **Angular, React, and Vue**. It has strong expertise in cloud platforms like **AWS, Azure, and GCP**, and uses data analytics tools such as **Power BI**, **Tableau**, and **Hadoop**. Infosys also excels in automation tools like **UiPath** and **BluePrism**, and advanced AI frameworks like **TensorFlow** and **PyTorch**. Employees continuously enhance their skills through the internal learning platform **Lex**.
Infosys is known for its disciplined, structured, and employee-friendly work culture. Freshers receive mentorship and opportunities for onsite projects abroad. The company follows a transparent placement process with an online assessment, technical interview, and HR interview. Candidates need 60% in academics with no active backlogs.

For communication, Infosys can be reached via email at **askus@infosys.com**, by phone at **+91-80-2852 0261**, or through their official LinkedIn page.

REPORTS OF PROJECT

EXISTING SYSTEM

When we Analysis the Manage about this firm then we face that they working with manual. And we all know that the manual system has many disadvantages.
Some are mentioned below…

- The manual system requires more time for processing.
- It requires more critical work.
- The manual system is more error prone.
- Difficult to maintain.
- Manual system is costly.
- Immediate response to the queries is difficult and time consuming.
- More men power needed.
- Manual system show of the particular place.

PROPOSED SYSTEM

- No Multiuser environment is provided. In this when one user is updating a form; the other user cannot open the same form to avoid conflicts.

- Security to the data is provided by means of Login Form. Only authorized users can have access to the system.
- The system allows users to maintain records of customers, suppliers, their orders, raw materials, stock availability as well as bill generation.
- This system also allows users to generate Customer reports, Supplier reports, sales and purchase reports as well as raw material's reports in the form of crystal reports.
- Easy to search and update Customer or supplier records by their Id.

SOFTWARE REQUIREMENTS SPECIFICATION

A **Dairy Management System** built using **C++** requires specific software and hardware components to ensure smooth operation. The software components provide the necessary tools for coding, compiling, and database management, while the hardware components ensure efficient performance in a real-world dairy store environment.

Software Requirements

1. Operating System (OS)

The Grocery Management System can run on multiple operating systems, including:

- **Windows 10/11** – Commonly used for development and deployment.
- **Linux (Ubuntu, Fedora, etc.)** – Preferred for server-based applications.
- **macOS** – Used by developers who prefer Mac for C++ programming.

2. Programming Language

- **C++** (Standard C++17 or later) is the core language used for developing the system.
- It provides high performance and efficient memory management, making it suitable for real-time applications.

3. Integrated Development Environment (IDE)

To write, compile, and debug C++ code, the following IDEs can be used:

- **Code::Blocks** – Lightweight and easy to use.
- **Dev-C++** – Suitable for beginners.
- **Microsoft Visual Studio** – Best for large-scale projects with GUI.
- **Qt Creator** – Ideal for GUI-based applications using the Qt framework.

4. Compiler

GCC (GNU Compiler Collection) – Used in Linux/macOS environments.
MinGW (Minimalist GNU for Windows) – Commonly used for Windows development.
MSVC (Microsoft Visual C++ Compiler) – Comes with Visual Studio.

5. Database Management System (DBMS)

A database is required to store product details, customer data, and sales records. Some commonly used databases are:
MySQL – Recommended for large-scale data handling.
SQLite – Lightweight, file-based database suitable for small businesses.
PostgreSQL – Advanced database with strong security features.
Libraries and Frameworks
Qt Framework – Used for building a graphical user interface (GUI).
MySQL Connector for C++ – Allows integration with MySQL databases.
Boost Library – Provides additional C++ functionalities.
OpenCV (Optional) – Can be used for barcode scanning or facial recognition.
Additional Software
XAMPP – A server environment for running MySQL databases locally.
Git – Used for version control and collaborative development.

HARDWARE REQUIREMENTS

Minimum Hardware Requirements

For basic functionality, the system can run on standard hardware with the following specifications:

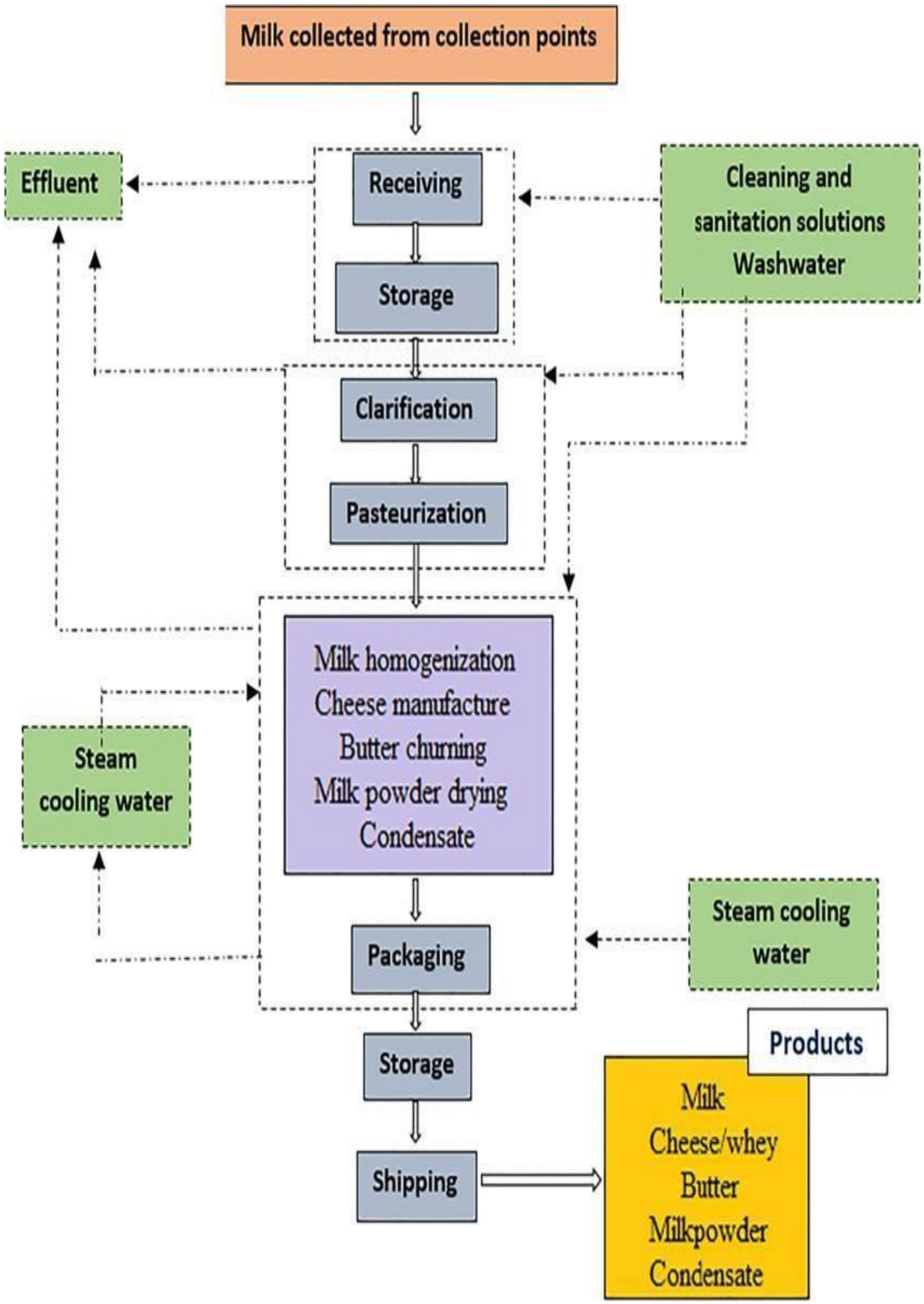
Component	Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3
RAM	4GB (Minimum) / 8GB (Recommended)
Storage	256GB HDD / 128GB SSD
Monitor	1366x768 resolution (Minimum)
Keyboard & Mouse	Standard input devices

Recommended Hardware for Large Grocery Stores

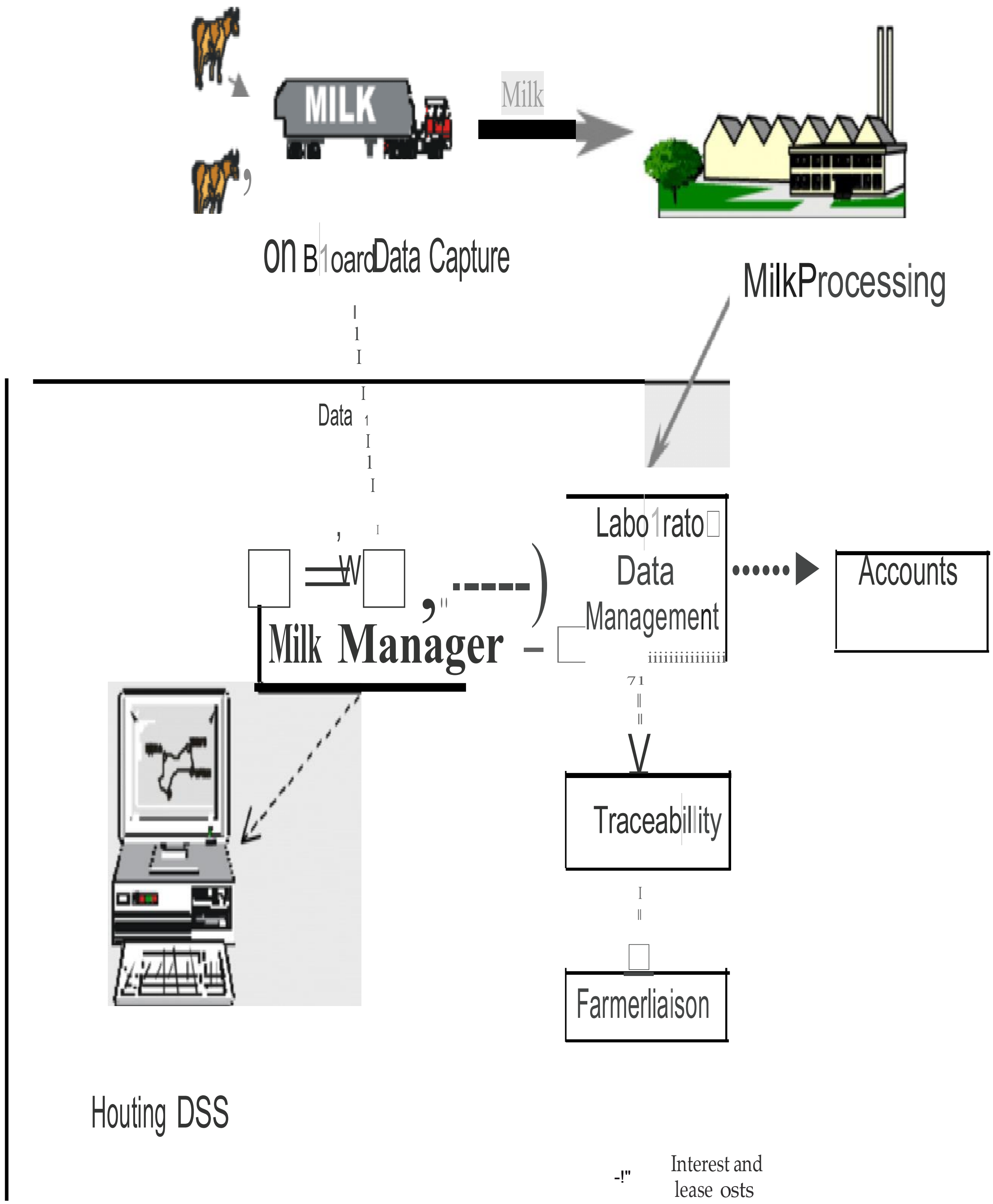
For businesses with high transaction volumes, a more powerful setup is needed

Component	Specification
Processor (CPU)	Intel Core i5/i7 or AMD Ryzen 5/7
RAM	8GB – 16GB
Storage	512GB SSD (For better speed)
Multi-Monitor Setup	Dual monitors (For cashier and admin use)
Touchscreen POS System	For easy billing and checkout
Barcode Scanner	USB/Handheld scanner for fast product entry
Thermal Receipt Printer	For generating customer bills
UPS Power (Uninterruptible Supply)	To prevent data loss during power failures

FLOWCHART OF DMS



STATE DIAGRAM OF DMS



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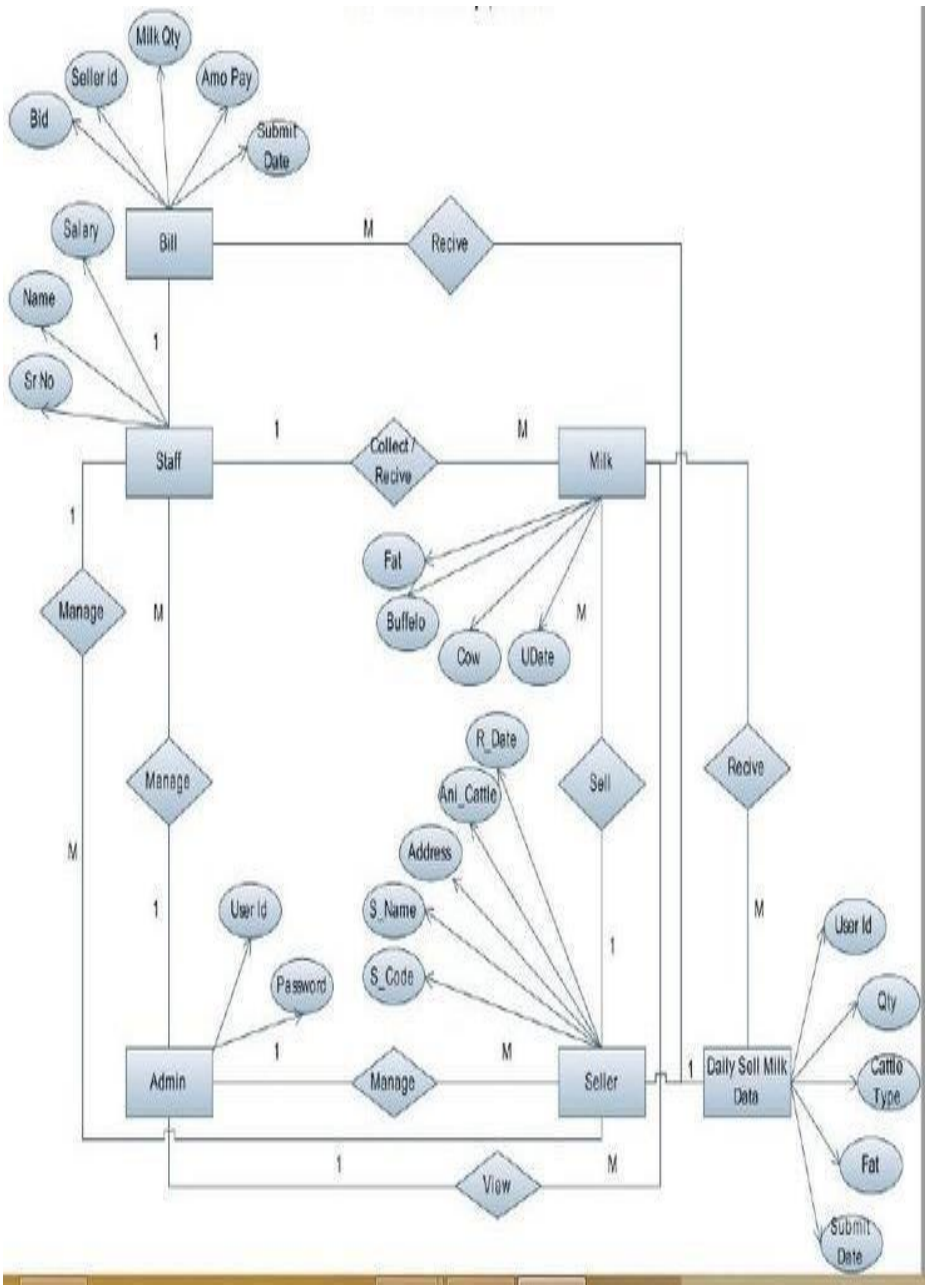


SYSTEM DESIGN OF DMS

ADMIN

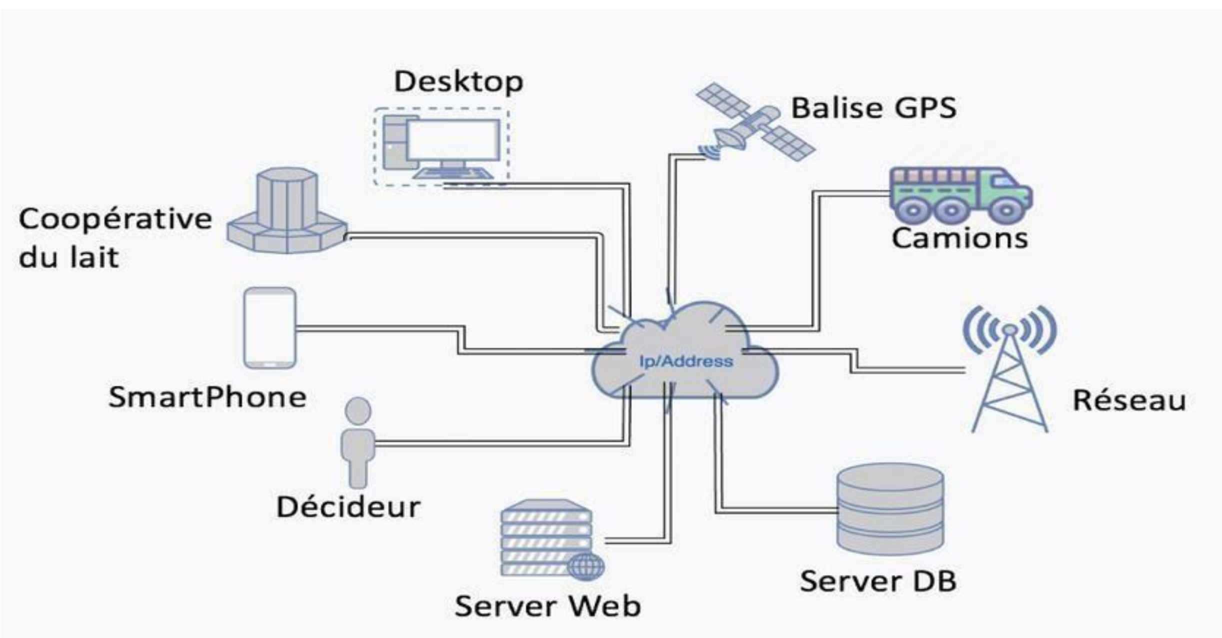
- Login
- Manages rates
- Manages staff
- Manages seller

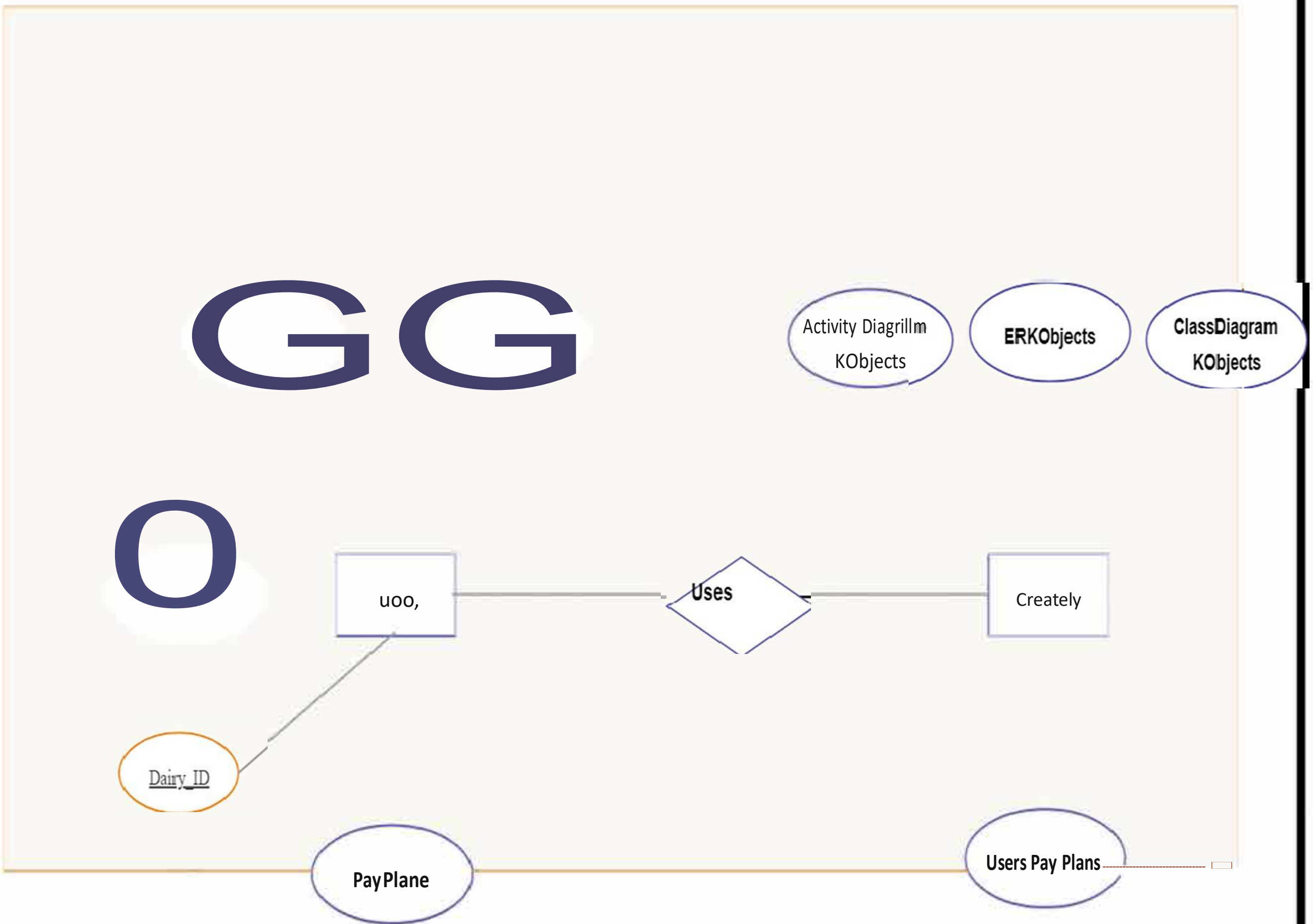
STAFF

- Login
- Seller Registration
- Manage feedback
- Insert milk details
- Seller
- Edit products
- Exit

USER

- Registration
- Login
- View rates
- Feedback

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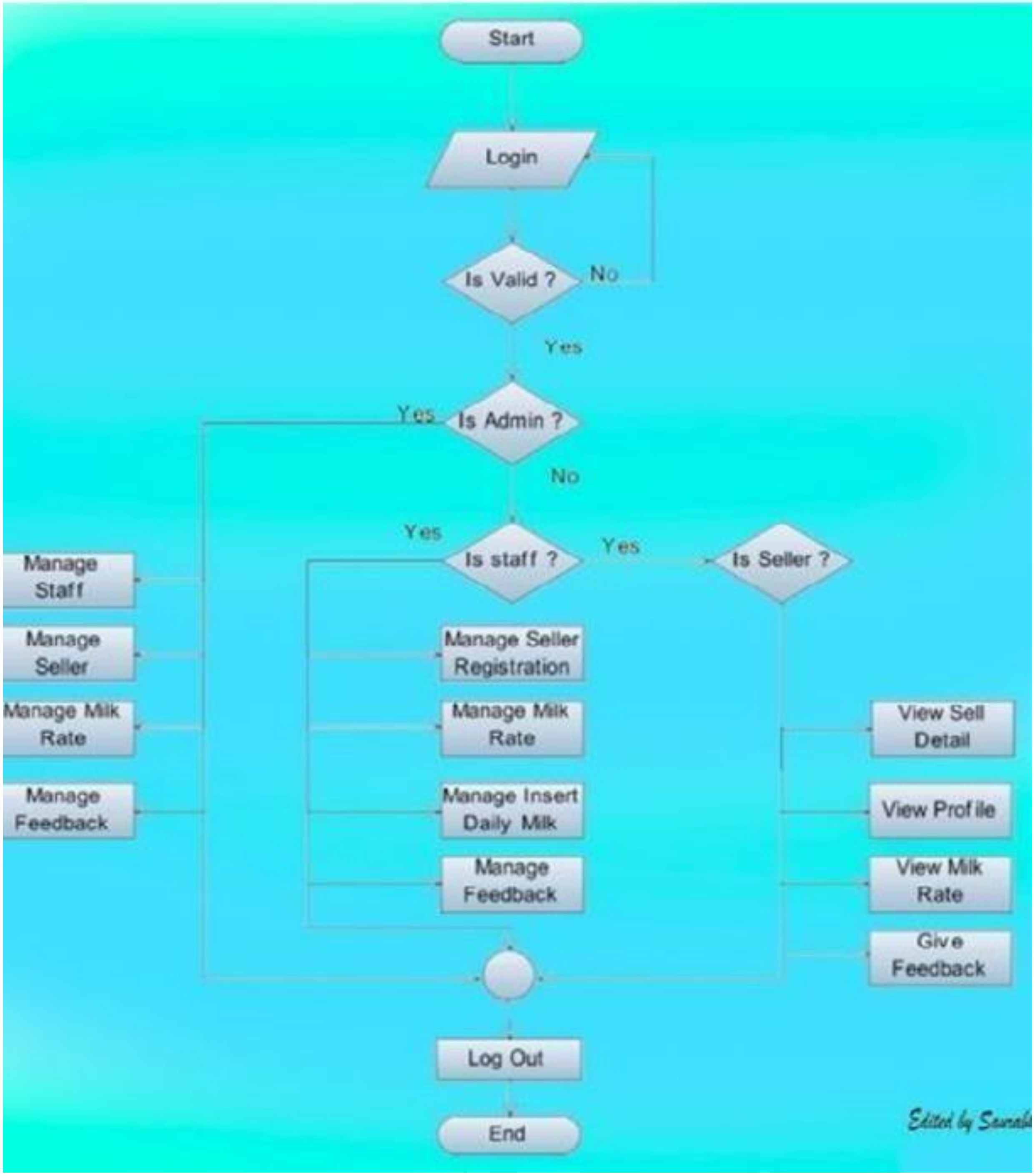
SYSTEMATIC FLOWCHART OF DMS



FEASIBILITY STUDY

A **feasibility study** evaluates whether the **Dairy Management System** is practical, cost-effective, and beneficial for implementation. It analyzes **technical, operational, economic, legal, and schedule feasibility** to determine the system's viability.

1. Technical Feasibility

Programming Language – The system is developed in **C++**, which is efficient, fast, and widely used.
File Handling – Uses **text files** to store inventory, billing, customer, and supplier data, eliminating the need for complex databases.
Hardware Requirements – Runs on any standard computer with basic specifications (Processor: **Intel i3+**, RAM: **2GB+**).
Software Requirements – Requires a **C++ compiler** (e.g., GCC, Turbo C++, or Microsoft Visual Studio).

2. Operational Feasibility

User-Friendly Interface – The system features a **simple menu- based interface**, making it easy for store employees to operate.
Efficient Inventory Management – Reduces manual work and human errors in stock tracking.
Fast & Accurate Billing – Ensures **quick calculations** and prevents overcharging or undercharging customers.
Data Storage & Retrieval – Stores customer and supplier records for easy future reference.

3.Economic Feasibility

- Low Development Cost – Built using C++, a free and open- source language, reducing licensing costs.
- No Need for Expensive Databases – Uses text files for data storage instead of paid database solutions.
- Minimal Hardware Investment – Runs on existing computers, avoiding additional hardware costs.
- Improves Business Efficiency – Saves time and labor costs by automating inventory and billing.

4. Schedule Feasibility

- Development Timeline – The project can be developed and implemented within 3–4 weeks.
- Easy Maintenance – Since it uses structured code and file handling, updates and bug fixes can be done quickly. Future Enhancements

5. Legal Feasibility

- **Compliance with Software Licensing** – Uses **C++, an open- source programming language**, ensuring no copyright issues.
- **Data Privacy & Security** – The system stores **basic customer and supplier details** without violating privacy laws.
- **Legally Acceptable Billing System** – Provides **accurate calculations and recordkeeping**, preventing fraud or tax-related issues.

6. Feasibility Study Conclusion

- The **Dairy Management System in C++** is **highly feasible** in all aspects:
- **Technically Feasible** – Runs on basic hardware and software.
- **Operationally Feasible** – Automates store operations, reducing errors and improving efficiency.
- **Economically Feasible** – Low-cost development with high business benefits.
- **Schedule Feasible** – Can be developed in a short period with room for future enhancements.
- **Legally Feasible** – Complies with licensing and privacy regulations.

SYSTEM ANALYSIS

Introduction

The **Dairy Management System** is designed to streamline the operations of a dairy store by automating inventory management, billing, and customer-supplier record keeping. It reduces manual errors, improves efficiency, and enhances business operations. This system analysis explores its **requirements, feasibility, components, and operational flow**.

Problem Statement

Managing a dairy store manually can lead to several challenges, including:
Inaccurate Inventory Tracking – Difficulty in keeping track of stock levels.
Billing Errors – Manual calculation mistakes can lead to financial losses.
Data Management Issues – Maintaining customer and supplier records manually is inefficient.
Time-Consuming Processes – Manual bookkeeping slows down operations.

System Objectives

Automate Inventory Management – Add, remove, and update items efficiently.
Generate Bills Instantly – Calculate total costs based on item price and quantity
Store Customer & Supplier Details – Maintain records for future reference.
Generate Sales Reports – Provide insights for business decision- making.

Feasibility Study

Technical Feasibility – The system is developed using **C++**, a widely used programming language, and uses **file handling for data storage**, making it suitable for small to medium businesses.

Operational Feasibility – The system is **user-friendly** with a simple menu-based interface that does not require extensive training.

Economic Feasibility – Since it is built using **open-source tools (C++, text files)**, there is **no extra cost** for implementation.

Schedule Feasibility – The system can be developed in **a few weeks**, making it a **quick solution** for store management.

System Components

User Interface – A command-line interface for users to interact with the system.
File Handling System – Used for storing inventory, customer, and supplier data.
Inventory Module – Allows adding, viewing, and updating grocery items.
Billing Module – Calculates bills and updates stock after purchases.
Report Generation Module – Generates reports for sales and inventory.
Customer & Supplier Management – Stores and retrieves customer/supplier details.

System Flowchart

1. User selects an option from the **menu**. 2. Based on selection:
Inventory is updated (Add/Delete/View items).

Billing is processed (Checks stock, calculates total, updates inventory).
Customer/Supplier details are stored/retrieved.
Sales reports are generated saves data to files for future use.

7. Advantages of the System

Eliminates manual errors in inventory and billing.
Fast and efficient – Saves time in store operations.
Easy to use – Simple interface requiring minimal training.
Data is securely stored in files for future reference.

8. Limitations & Future Enhancements

No GUI – Can be improved using **Qt or Tkinter**.
No database – Can integrate **MySQL or SQLite** for better data management.
No multi-user access – Can be expanded for **cloud-based use**.
No real-time tracking – Can integrate **barcode scanning and AI- based forecasting**.

TESTING

1. FUNCTIONAL TESTING

Focuses on checking individual features to ensure they work as intended

Test Case ID	Feature	Test Description	Expected Outcome	Status
TC001	Add Item	Add a valid item to the inventory.	Item appears in inventory.txt with correct details.	
TC002	Add Item	Add an item with a negative price or quantity.	System rejects invalid input.	
TC003	Display Items	Display inventory with multiple items.	All items appear with proper formatting.	
TC004	Generate Bill	Purchase valid item with available stock.	Bill generated with correct total. Stock reduces accordingly.	
TC005	Generate Bill	Attempt to purchase an item that doesn't exist.	Error message: "Item not found!"	

TC006	Generate Bill	Attempt to purchase an item with insufficient stock.	Error message: "Not enough stock!"
TC007	Add Customer	Add a customer with a valid phone number.	Customer saved in customers.txt.
TC008	Add Customer	Enter an invalid phone number (less than 10 digits).	Error message: "Invalid phone number!"
TC009	Add Supplier	Add a valid supplier with a contact number.	Supplier appears in suppliers.txt.
TC010	Exit System	Select exit option.	Program terminates gracefully.

FUTURE SCOPE

The Typing Speed Tester project provides a foundation for improving typing skills, tracking performance, and encouraging healthy competition. In the future, the project can be enhanced with the following features and improvements:

1. **Graphical User Interface (GUI) Enhancement**
   - Transition from a console-based interface to a user-friendly GUI using Tkinter, PyQt, Enhancement
   - Include interactive buttons, colorful progress bars, and visual feedback for errors and performance.
2. **Customizable Typing Tests**
   - Allow users to select difficulty levels, test durations, and themes (e.g., coding text, general knowledge, quotes).
   - Include adaptive tests that adjust difficulty based on user performance.
3. **Mobile or Web Version**
   - Develop the application as a web app or mobile app to make it accessible on multiple platforms.
   - Enable cloud-based storage for user data and leaderboards.
4. **Integration of Gamification Elements**
   - Introduce badges, achievements, and levels to motivate users.
   - Add time-based challenges, streaks, and rewards for continuous improvement.
5. **Support for Multiple Languages**
   - Allow typing tests in different languages to cater to a wider user base.
6. Include bilingual or multilingual typing practice options.
7. **Voice Typing and AI Assistance (Advanced)**
   - Incorporate voice-to-text typing for training speed and accuracy.
   - Use AI to suggest tips for improving typing style and reducing errors.
8. Include bilingual or multilingual typing practice options.
9. **Voice Typing and AI Assistance (Advanced)**
   - Incorporate voice-to-text typing for training speed and accuracy.
   - Use AI to suggest tips for improving typing style and reducing errors.

10. Include bilingual or multilingual typing practice options.

11. Voice Typing and AI Assistance (Advanced)

- Incorporate voice-to-text typing for training speed and accuracy.
- Use AI to suggest tips for improving typing style and reducing errors.

Conclusion

The Dairy Management System is designed to simplify and automate the daily operations of a dairy business. The system helps in maintaining records of suppliers, customers, milk purchase, sales, and billing efficiently. By replacing the manual system, it reduces human errors, saves time, and improves overall productivity. The project demonstrates how software technology can make dairy management more organized, accurate, and user-friendly. In the future, the system can be enhanced with a graphical user interface, database integration, and online access to make it more powerful and suitable for real-world dairy industries.

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